

Robert Lee

Senior Software Engineer

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Microsoft engineer building **billion-scale retrieval systems** for enterprise AI. Technical leadership across vector search, agentic retrieval, and distributed systems, from research and architecture through production launch and reliability.

Experience

Senior Software Engineer

Mar. 2025 – Present

Microsoft — Azure AI Search

Redmond, WA

- Engineering lead for a **vector-search diversity capability**, owning scoping, design, and implementation. Adapted Microsoft Research work to broaden evidence for complex queries at **billions-of-vectors scale**.
- Led **agentic filter and boost generation** from research to production. Reframed unbounded synthesis as a bounded, verifiable operator set; validated coverage against customer usage, aligned research and product teams, and shipped improvements to Public Preview.
- Architected agentic-retrieval benchmarking across CPU, memory, throughput, and latency; proposed and shipped the **production billing model** for a new serverless enterprise search offering.
- Integrated retrieval into tool-calling and multi-agent workflows, grounding LLM agents in governed enterprise knowledge for retrieval-augmented generation (RAG).
- Lead cross-team architecture reviews and technical decisions; mentor and onboard engineers across information retrieval, vector search, Azure OpenAI, and agent orchestration.

Software Engineer II

Jun. 2022 – Feb. 2025

Microsoft — Azure AI Search

Redmond, WA

- Built and scaled vector retrieval capabilities across billions of embeddings. Drove **vector quantization from Public Preview to GA**, delivering **8–32x customer cost savings** and **up to 20x lower latency** through scalar and binary representations and efficient distance computation.
- Optimized vector distance kernels with **SIMD, loop unrolling, independent accumulators, and fused multiply-add**, improving instruction-level parallelism and throughput.
- Designed hybrid-search subscore fusion and score thresholding to improve relevance across blended vector and keyword retrieval.
- Designed resource-aware quota enforcement for HNSW indexes, **reducing limit overshoot by 100x** through cross-team design collaboration.
- Owned an **extensible facet-aggregation engine** from specification through delivery, implementing a lexer, parser, and evaluator with BNF grammar, shunting-yard parsing, and Reverse Polish Notation.
- Root-caused cross-team production incidents, caught a critical quantization bug before release, and reviewed distributed-system and vector-algorithm changes for correctness and reliability.

Software Engineer

Jul. 2021 – May 2022

Microsoft — Azure AI Search

Redmond, WA

- Delivered index aliases to Public Preview, allowing applications to switch search indexes with **no code changes or downtime**.
- Completed a phased, zero-downtime telemetry-table migration that preserved data integrity and produced a **50–100x query speedup**.

Technical Skills

Languages: C++, C#, Java, Python, TypeScript.

Search & AI: Vector search, quantization, hybrid retrieval, RAG, LLM agents, graph search, computer vision, TensorFlow, Keras.

Systems: Distributed and durable systems, SIMD optimization, benchmarking, authentication, production diagnostics, Git, CI.

Earlier Microsoft Experience

- **Software Engineer Intern, Azure Cognitive Search** (Sep.–Dec. 2020): shipped the team’s second most requested API and backend feature in C# to preview.
- **Software Engineer Intern, Azure Search** (Jun.–Aug. 2019): built a TypeScript search-site generator, query suggestions, filtering, and portal editing tools.
- **Software Developer Intern, Microsoft Garage** (Jan.–Apr. 2018): built an offline chest X-ray classification app, image pipeline, CI, and mobile TensorFlow integration.

Selected Machine Learning Projects

Human Pose Estimation — Team Lead Jan.–May 2021

- Led development of a heatmap-based 2D pose model on COCO 2017, **trained from scratch**. Architected the cloud training pipeline, model, augmentation, visualization, and deployment.

Semantic Road Segmentation Jul.–Sep. 2020

- Built a full U-Net training pipeline on KITTI Road with Keras, custom losses, augmentation, and cloud training; achieved **up to 99.1% F1**, with 91% in the worst case.

Self-Supervised Monocular Depth Estimation Jan.–May 2020

- Adapted Monodepth2 to TensorFlow and **136,000 DrivingStereo image pairs**, using stereo reconstruction, multi-scale loss, and auto-masking without ground-truth depth.

Leadership & Community

- **Industry Mentor, University of Victoria** (Sep.–Dec. 2024; Aug. 2026–Present): authored challenges in LLM reliability and production AI; coach student teams on requirements, scope, technical assumptions, and evaluation.
- **Founder & Director, Seniors’ Digital Literacy Program** (2015–2021): grew the program to **180 volunteers and 650+ seniors across 30 workshops**; secured institutional support and built a succession plan that sustained it beyond my tenure.
- **IEEE Student Branch Chair & Vice-Chair** (2019–2021): co-led a six-person executive; built and taught a **14-workshop series reaching 350+ students**.
- **Founder & Chair, UVic Tech & Business Strategy Conference** (2019): led planning, speakers, and execution for **200+ attendees**.

Education & Recognition

B.Eng., Electrical & Computer Engineering 2016–2021

University of Victoria — **97% cumulative average**.

Undergraduate research: hardware acceleration for machine-learning neural networks.

Schulich Leader Scholarship: \$80,000 full ride; one of 50 awarded nationally (2016). **National Champion, Michael Smith Science Challenge:** record 97.5% among 1,700+ candidates (2014). **Western Engineering Competition:** 1st, Senior Design (2020). **Y Combinator AI Startup School:** selected for the inaugural cohort (2025).